

Feature Engine Pro

Automated Feature Selection - Audit Report

Executive Summary

Generated on 2026-05-05 11:14:16 • Pipeline runtime: 20.97s

93

INPUT FEATURES

4

SELECTED FEATURES

89

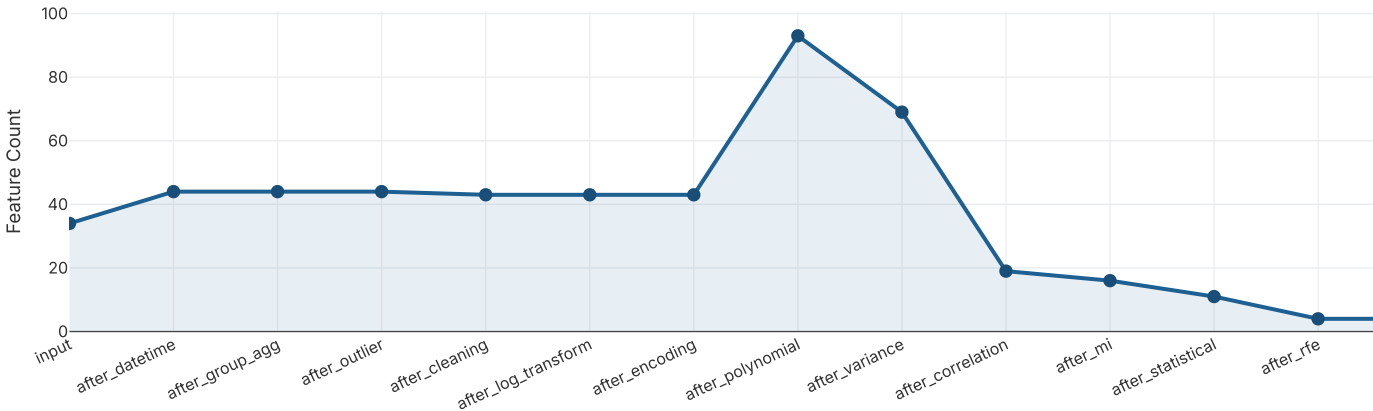
FEATURES DROPPED

95.7%

DIMENSIONALITY
REDUCTION

Column Journey

Feature count at each pipeline stage — shows where features were created and destroyed



Performance Impact

Proving that fewer features = same or better performance

93

FEATURES BEFORE

4

FEATURES AFTER

95.7%

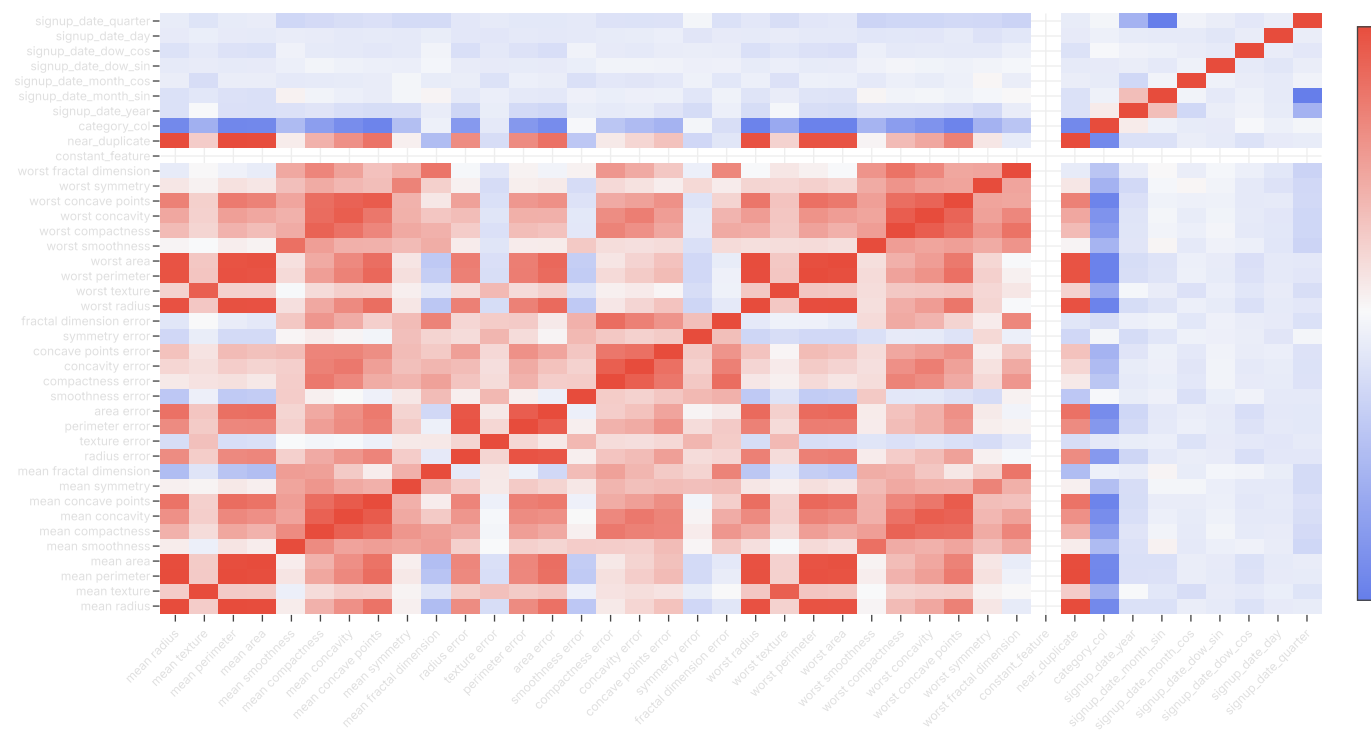
REDUCTION

Metric	Before Selection	After Selection	Change
accuracy	0.9978 ± 0.0044	0.9978 ± 0.0044	+ 0.0000
f1_weighted	0.9978 ± 0.0044	0.9978 ± 0.0044	+ 0.0000



Correlation Matrix

Interactive heatmap — hover for details, scroll to zoom



The Audit Trail

Complete history of feature transformations and selection reasoning

Search features, steps, or reasons...

- All Features (93)
- Kept (4)
- Dropped (89)

Feature	Final Status	Pipeline Step	Action & Reasoning
mean radius	✓ KEPT	VarianceThreshold	Variance 11.065510 above threshold 0.01.
mean texture	✓ KEPT	VarianceThreshold	Variance 16.572300 above threshold 0.01.
mean perimeter	✓ KEPT	VarianceThreshold	Variance 523.432372 above threshold 0.01.
mean area	✓ KEPT	VarianceThreshold	Variance 90536.810847 above threshold 0.01.
mean smoothness	✗ DROPPED	VarianceThreshold	Variance 0.000180 below threshold 0.01.
mean compactness	✗ DROPPED	VarianceThreshold	Variance 0.002503 below threshold 0.01.
mean concavity	✗ DROPPED	VarianceThreshold	Variance 0.004361 below threshold 0.01.
mean concave points	✗ DROPPED	VarianceThreshold	Variance 0.001355 below threshold 0.01.
mean symmetry	✗ DROPPED	VarianceThreshold	Variance 0.000687 below threshold 0.01.
mean fractal dimension	✗ DROPPED	VarianceThreshold	Variance 0.000042 below threshold 0.01.
radius error	✓ KEPT	VarianceThreshold	Variance 0.018058 above threshold 0.01.
texture error	✓ KEPT	VarianceThreshold	Variance 0.219989 above threshold 0.01.
perimeter error	✓ KEPT	VarianceThreshold	Variance 1.767382 above threshold 0.01.
area error	✓ KEPT	VarianceThreshold	Variance 0.367620 above threshold 0.01.
smoothness error	✗ DROPPED	VarianceThreshold	Variance 0.000006 below threshold 0.01.
compactness error	✗ DROPPED	VarianceThreshold	Variance 0.000230 below threshold 0.01.

Feature	Final Status	Pipeline Step	Action & Reasoning
concavity error	X DROPPED	VarianceThreshold	Variance 0.000443 below threshold 0.01.
concave points error	X DROPPED	VarianceThreshold	Variance 0.000030 below threshold 0.01.
symmetry error	X DROPPED	VarianceThreshold	Variance 0.000042 below threshold 0.01.
fractal dimension error	X DROPPED	VarianceThreshold	Variance 0.000003 below threshold 0.01.
worst radius	✓ KEPT	VarianceThreshold	Variance 20.190760 above threshold 0.01.
worst texture	✓ KEPT	VarianceThreshold	Variance 35.749389 above threshold 0.01.
worst perimeter	✓ KEPT	VarianceThreshold	Variance 1003.810023 above threshold 0.01.
worst area	✓ KEPT	VarianceThreshold	Variance 205069.400445 above threshold 0.01.
worst smoothness	X DROPPED	VarianceThreshold	Variance 0.000518 below threshold 0.01.
worst compactness	✓ KEPT	VarianceThreshold	Variance 0.020652 above threshold 0.01.
worst concavity	✓ KEPT	VarianceThreshold	Variance 0.039708 above threshold 0.01.
worst concave points	X DROPPED	VarianceThreshold	Variance 0.004267 below threshold 0.01.
worst symmetry	X DROPPED	VarianceThreshold	Variance 0.002975 below threshold 0.01.
worst fractal dimension	X DROPPED	VarianceThreshold	Variance 0.000266 below threshold 0.01.
constant_feature	X DROPPED	VarianceThreshold	Quasi-constant: 100.0% of values are '5'. Threshold: 99%.
near_duplicate	✓ KEPT	VarianceThreshold	Variance 11.067723 above threshold 0.01.
category_col	✓ KEPT	VarianceThreshold	Binary feature protected. Variance: 0.201940.
signup_date_year	✓ KEPT	VarianceThreshold	Binary feature protected. Variance: 0.229143.
signup_date_month_sin	✓ KEPT	VarianceThreshold	Variance 0.459879 above threshold 0.01.

Feature	Final Status	Pipeline Step	Action & Reasoning
signup_date_month_cos	✓ KEPT	VarianceThreshold	Variance 0.504467 above threshold 0.01.
signup_date_dow_sin	✓ KEPT	VarianceThreshold	Variance 0.506983 above threshold 0.01.
signup_date_dow_cos	✓ KEPT	VarianceThreshold	Variance 0.494847 above threshold 0.01.
signup_date_day	✓ KEPT	VarianceThreshold	Variance 77.493373 above threshold 0.01.
signup_date_quarter	✓ KEPT	VarianceThreshold	Variance 1.151745 above threshold 0.01.
signup_date_is_weekend	✓ KEPT	VarianceThreshold	Binary feature protected. Variance: 0.205470.
signup_date_is_month_start	X DROPPED	VarianceThreshold	Quasi-constant: 100.0% of values are '0'. Threshold: 99%.
signup_date_days_since_epoch	✓ KEPT	VarianceThreshold	Variance 26010.743506 above threshold 0.01.
category_col_x_worst perimeter	✓ KEPT	VarianceThreshold	Variance 1652.233058 above threshold 0.01.
category_col_x_worst area	✓ KEPT	VarianceThreshold	Variance 85599.213115 above threshold 0.01.
category_col_x_mean concave points	X DROPPED	VarianceThreshold	Variance 0.000333 below threshold 0.01.
category_col_x_worst concave points	X DROPPED	VarianceThreshold	Variance 0.002029 below threshold 0.01.
category_col_x_worst radius	✓ KEPT	VarianceThreshold	Variance 38.643631 above threshold 0.01.
category_col_x_mean perimeter	✓ KEPT	VarianceThreshold	Variance 1312.142125 above threshold 0.01.
category_col_x_mean area	✓ KEPT	VarianceThreshold	Variance 56319.046953 above threshold 0.01.
category_col_x_mean radius	✓ KEPT	VarianceThreshold	Variance 31.617798 above threshold 0.01.
category_col_x_near_duplicate	✓ KEPT	VarianceThreshold	Variance 31.624122 above threshold 0.01.
category_col_x_mean concavity	X DROPPED	VarianceThreshold	Variance 0.001482 below threshold 0.01.
category_col_x_area error	✓ KEPT	VarianceThreshold	Variance 1.932461 above threshold 0.01.

Feature	Final Status	Pipeline Step	Action & Reasoning
category_col__x__worst concavity	✓ KEPT	VarianceThreshold	Variance 0.018175 above threshold 0.01.
category_col__x__perimeter error	✓ KEPT	VarianceThreshold	Variance 1.208091 above threshold 0.01.
category_col__x__radius error	✓ KEPT	VarianceThreshold	Variance 0.016581 above threshold 0.01.
worst perimeter__x__worst area	✓ KEPT	VarianceThreshold	Variance 7824715332.182327 above threshold 0.01.
worst perimeter__x__mean concave points	✓ KEPT	VarianceThreshold	Variance 40.378982 above threshold 0.01.
worst perimeter__x__worst concave points	✓ KEPT	VarianceThreshold	Variance 130.874425 above threshold 0.01.
worst perimeter__x__worst radius	✓ KEPT	VarianceThreshold	Variance 1241517.001664 above threshold 0.01.
worst perimeter__x__mean perimeter	✓ KEPT	VarianceThreshold	Variance 34613709.129448 above threshold 0.01.
worst perimeter__x__mean area	✓ KEPT	VarianceThreshold	Variance 3642029816.206515 above threshold 0.01.
worst perimeter__x__mean radius	✓ KEPT	VarianceThreshold	Variance 766205.170934 above threshold 0.01.
worst perimeter__x__near_duplicate	✓ KEPT	VarianceThreshold	Variance 766358.419630 above threshold 0.01.
worst perimeter__x__mean concavity	✓ KEPT	VarianceThreshold	Variance 111.917032 above threshold 0.01.
worst perimeter__x__area error	✓ KEPT	VarianceThreshold	Variance 31784.459223 above threshold 0.01.
worst perimeter__x__worst concavity	✓ KEPT	VarianceThreshold	Variance 922.649144 above threshold 0.01.
worst perimeter__x__perimeter error	✓ KEPT	VarianceThreshold	Variance 63492.197319 above threshold 0.01.
worst perimeter__x__radius error	✓ KEPT	VarianceThreshold	Variance 693.981544 above threshold 0.01.
worst area__x__mean concave points	✓ KEPT	VarianceThreshold	Variance 4316.715905 above threshold 0.01.
worst area__x__worst concave points	✓ KEPT	VarianceThreshold	Variance 14470.246843 above threshold 0.01.
worst area__x__worst radius	✓ KEPT	VarianceThreshold	Variance 168890039.271614 above threshold 0.01.

Feature	Final Status	Pipeline Step	Action & Reasoning
worst area__x__mean perimeter	✓ KEPT	VarianceThreshold	Variance 4785743618.220289 above threshold 0.01.
worst area__x__mean area	✓ KEPT	VarianceThreshold	Variance 449587964340.562317 above threshold 0.01.
worst area__x__mean radius	✓ KEPT	VarianceThreshold	Variance 107720210.340526 above threshold 0.01.
worst area__x__near_duplicate	✓ KEPT	VarianceThreshold	Variance 107741755.459797 above threshold 0.01.
worst area__x__mean concavity	✓ KEPT	VarianceThreshold	Variance 11461.788937 above threshold 0.01.
category_col__squared	✓ KEPT	VarianceThreshold	Binary feature protected. Variance: 0.204848.
worst perimeter__squared	✓ KEPT	VarianceThreshold	Variance 59138922.988873 above threshold 0.01.
worst area__squared	✓ KEPT	VarianceThreshold	Variance 948646614563.955444 above threshold 0.01.
mean concave points__squared	X DROPPED	VarianceThreshold	Variance 0.000026 below threshold 0.01.
worst concave points__squared	X DROPPED	VarianceThreshold	Variance 0.000315 below threshold 0.01.
worst radius__squared	✓ KEPT	VarianceThreshold	Variance 26362.405267 above threshold 0.01.
mean perimeter__squared	✓ KEPT	VarianceThreshold	Variance 20962505.653115 above threshold 0.01.
mean area__squared	✓ KEPT	VarianceThreshold	Variance 222691139018.883575 above threshold 0.01.
mean radius__squared	✓ KEPT	VarianceThreshold	Variance 10272.947268 above threshold 0.01.
near_duplicate__squared	✓ KEPT	VarianceThreshold	Variance 10277.057063 above threshold 0.01.
mean concavity__squared	X DROPPED	VarianceThreshold	Variance 0.000237 below threshold 0.01.
area error__squared	✓ KEPT	VarianceThreshold	Variance 17.901149 above threshold 0.01.
worst concavity__squared	✓ KEPT	VarianceThreshold	Variance 0.021439 above threshold 0.01.

Feature	Final Status	Pipeline Step	Action & Reasoning
perimeter error__squared	✓ KEPT	VarianceThreshold	Variance 80.527098 above threshold 0.01.
radius error__squared	✓ KEPT	VarianceThreshold	Variance 0.010010 above threshold 0.01.
mean radius	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.97). Kept worst perimeter (higher target corr: 0.787 vs 0.719).
mean perimeter	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.98). Kept worst perimeter (higher target corr: 0.787 vs 0.735).
mean area	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.97). Kept worst perimeter (higher target corr: 0.787 vs 0.720).
worst radius	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.778).
worst perimeter	✓ KEPT	CorrelationSelector	Best representative in cluster of 26 features. Target correlation: 0.787.
worst area	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.777).
near_duplicate	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.97). Kept worst perimeter (higher target corr: 0.787 vs 0.719).
worst perimeter__x__worst area	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 1.00). Kept worst perimeter (higher target corr: 0.787 vs 0.783).
worst perimeter__x__worst radius	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 1.00). Kept worst perimeter (higher target corr: 0.787 vs 0.784).
worst perimeter__x__mean perimeter	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 1.00). Kept worst perimeter (higher target corr: 0.787 vs 0.769).
worst perimeter__x__mean area	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.751).
worst perimeter__x__mean radius	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.764).

Feature	Final Status	Pipeline Step	Action & Reasoning
worst perimeter__x__near_duplicate	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.764).
worst perimeter__x__area error	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.96). Kept worst perimeter (higher target corr: 0.787 vs 0.783).
worst area__x__worst radius	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.777).
worst area__x__mean perimeter	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.768).
worst area__x__mean area	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.755).
worst area__x__mean radius	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.763).
worst area__x__near_duplicate	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.763).
worst perimeter__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 1.00). Kept worst perimeter (higher target corr: 0.787 vs 0.787).
worst area__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.777).
worst radius__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.99). Kept worst perimeter (higher target corr: 0.787 vs 0.778).
mean perimeter__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.98). Kept worst perimeter (higher target corr: 0.787 vs 0.735).
mean area__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.97). Kept worst perimeter (higher target corr: 0.787 vs 0.720).
mean radius__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.97). Kept worst perimeter (higher target corr: 0.787 vs 0.719).
near_duplicate__squared	X DROPPED	CorrelationSelector	Clustered with 26 features (corr ≈ 0.97). Kept worst perimeter (higher target corr: 0.787 vs 0.719).

Feature	Final Status	Pipeline Step	Action & Reasoning
			0.787 vs 0.719).
mean texture	X DROPPED	CorrelationSelector	Clustered with 2 features (corr $\approx$ 0.90). Kept worst texture (higher target corr: 0.491 vs 0.480).
worst texture	✓ KEPT	CorrelationSelector	Best representative in cluster of 2 features. Target correlation: 0.491.
radius error	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.93). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.589).
perimeter error	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.92). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.607).
area error	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.99). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.694).
worst perimeter__x__perimeter error	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.98). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.727).
worst perimeter__x__radius error	✓ KEPT	CorrelationSelector	Best representative in cluster of 8 features. Target correlation: 0.729.
area error__squared	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.99). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.694).
perimeter error__squared	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.92). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.607).
radius error__squared	X DROPPED	CorrelationSelector	Clustered with 8 features (corr $\approx$ 0.93). Kept worst perimeter__x__radius error (higher target corr: 0.729 vs 0.589).
texture error	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
worst compactness	X DROPPED	CorrelationSelector	Clustered with 4 features (corr $\approx$ 0.88). Kept worst perimeter__x__worst concavity (higher target corr: 0.767 vs 0.599).
worst concavity	X DROPPED	CorrelationSelector	Clustered with 4 features (corr $\approx$ 0.98). Kept worst perimeter__x__worst concavity (higher target corr: 0.767 vs 0.698).

Feature	Final Status	Pipeline Step	Action & Reasoning
worst_perimeter_x_worst_concavity	✓ KEPT	CorrelationSelector	Best representative in cluster of 4 features. Target correlation: 0.767.
worst_concavity_squared	X DROPPED	CorrelationSelector	Clustered with 4 features (corr ≈ 0.98). Kept worst_perimeter_x_worst_concavity (higher target corr: 0.767 vs 0.698).
category_col	✓ KEPT	CorrelationSelector	Best representative in cluster of 2 features. Target correlation: 0.977.
category_col_squared	X DROPPED	CorrelationSelector	Clustered with 2 features (corr ≈ 1.00). Kept category_col (higher target corr: 0.977 vs 0.977).
signup_date_year	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_month_sin	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_month_cos	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_dow_sin	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_dow_cos	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_day	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_quarter	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_is_weekend	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
signup_date_days_since_epoch	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
category_col_x_worst_perimeter	X DROPPED	CorrelationSelector	Clustered with 7 features (corr ≈ 0.98). Kept category_col_x_mean_area (higher target corr: 0.799 vs 0.791).
category_col_x_worst_area	X DROPPED	CorrelationSelector	Clustered with 7 features (corr ≈ 0.99). Kept category_col_x_mean_area (higher target corr: 0.799 vs 0.793).
category_col_x_worst_radius	X DROPPED	CorrelationSelector	Clustered with 7 features (corr ≈ 0.99). Kept category_col_x_mean_area (higher target corr: 0.799 vs 0.791).

Feature	Final Status	Pipeline Step	Action & Reasoning
category_col_x_mean perimeter	X DROPPED	CorrelationSelector	Clustered with 7 features (corr $\approx$ 1.00). Kept category_col_x_mean area (higher target corr: 0.799 vs 0.796).
category_col_x_mean area	✓ KEPT	CorrelationSelector	Best representative in cluster of 7 features. Target correlation: 0.799.
category_col_x_mean radius	X DROPPED	CorrelationSelector	Clustered with 7 features (corr $\approx$ 1.00). Kept category_col_x_mean area (higher target corr: 0.799 vs 0.799).
category_col_x_near_duplicate	X DROPPED	CorrelationSelector	Clustered with 7 features (corr $\approx$ 1.00). Kept category_col_x_mean area (higher target corr: 0.799 vs 0.799).
category_col_x_area error	✓ KEPT	CorrelationSelector	Best representative in cluster of 3 features. Target correlation: 0.792.
category_col_x_perimeter error	X DROPPED	CorrelationSelector	Clustered with 3 features (corr $\approx$ 0.96). Kept category_col_x_area error (higher target corr: 0.792 vs 0.792).
category_col_x_radius error	X DROPPED	CorrelationSelector	Clustered with 3 features (corr $\approx$ 0.98). Kept category_col_x_area error (higher target corr: 0.792 vs 0.791).
category_col_x_worst concavity	✓ KEPT	CorrelationSelector	No collinearity cluster (unique group). Method: spearman.
worst_perimeter_x_mean_concave_points	X DROPPED	CorrelationSelector	Clustered with 6 features (corr $\approx$ 0.96). Kept worst_area_x_worst_concave_points (higher target corr: 0.809 vs 0.797).
worst_perimeter_x_worst_concave_points	X DROPPED	CorrelationSelector	Clustered with 6 features (corr $\approx$ 0.99). Kept worst_area_x_worst_concave_points (higher target corr: 0.809 vs 0.800).
worst_perimeter_x_mean_concavity	X DROPPED	CorrelationSelector	Clustered with 6 features (corr $\approx$ 0.91). Kept worst_area_x_worst_concave_points (higher target corr: 0.809 vs 0.772).
worst_area_x_mean_concave_points	X DROPPED	CorrelationSelector	Clustered with 6 features (corr $\approx$ 0.97). Kept worst_area_x_worst_concave_points (higher target corr: 0.809 vs 0.806).
worst_area_x_worst_concave_points	✓ KEPT	CorrelationSelector	Best representative in cluster of 6 features. Target correlation: 0.809.
worst_area_x_mean_concavity	X DROPPED	CorrelationSelector	Clustered with 6 features (corr $\approx$ 0.95). Kept worst_area_x_worst_concave

Feature	Final Status	Pipeline Step	Action & Reasoning
			points (higher target corr: 0.809 vs 0.795).
texture error	X DROPPED	MutualInformation	MI = 0.0012 ± 0.0017 < threshold 0.0100. Weak non-linear dependency.
worst texture	✓ KEPT	MutualInformation	MI = 0.1498 ± 0.0016 ≥ threshold 0.0100. (3 runs averaged).
worst perimeter	✓ KEPT	MutualInformation	MI = 0.4661 ± 0.0008 ≥ threshold 0.0100. (3 runs averaged).
category_col	✓ KEPT	MutualInformation	MI = 0.6031 ± 0.0042 ≥ threshold 0.0100. (3 runs averaged).
signup_date_year	✓ KEPT	MutualInformation	MI = 0.0183 ± 0.0000 ≥ threshold 0.0100. (3 runs averaged).
signup_date_month_sin	✓ KEPT	MutualInformation	MI = 0.0221 ± 0.0052 ≥ threshold 0.0100. (3 runs averaged).
signup_date_month_cos	✓ KEPT	MutualInformation	MI = 0.0115 ± 0.0145 ≥ threshold 0.0100. (3 runs averaged).
signup_date_dow_sin	✓ KEPT	MutualInformation	MI = 0.0177 ± 0.0161 ≥ threshold 0.0100. (3 runs averaged).
signup_date_dow_cos	X DROPPED	MutualInformation	MI = 0.0013 ± 0.0018 < threshold 0.0100. Weak non-linear dependency.
signup_date_day	✓ KEPT	MutualInformation	MI = 0.0511 ± 0.0212 ≥ threshold 0.0100. (3 runs averaged).
signup_date_quarter	✓ KEPT	MutualInformation	MI = 0.0162 ± 0.0000 ≥ threshold 0.0100. (3 runs averaged).
signup_date_is_weekend	X DROPPED	MutualInformation	MI = 0.0004 ± 0.0000 < threshold 0.0100. Weak non-linear dependency.
signup_date_days_since_epoch	✓ KEPT	MutualInformation	MI = 0.0983 ± 0.0011 ≥ threshold 0.0100. (3 runs averaged).
category_col_x_mean area	✓ KEPT	MutualInformation	MI = 0.6067 ± 0.0000 ≥ threshold 0.0100. (3 runs averaged).
category_col_x_area error	✓ KEPT	MutualInformation	MI = 0.6128 ± 0.0000 ≥ threshold 0.0100. (3 runs averaged).
category_col_x_worst concavity	✓ KEPT	MutualInformation	MI = 0.4691 ± 0.0000 ≥ threshold 0.0100. (3 runs averaged).
worst perimeter_x_worst concavity	✓ KEPT	MutualInformation	MI = 0.4194 ± 0.0000 ≥ threshold 0.0100. (3 runs averaged).

Feature	Final Status	Pipeline Step	Action & Reasoning
worst perimeter__x__radius error	✓ KEPT	MutualInformation	MI = $0.3625 \pm 0.0000 \geq$ threshold 0.0100. (3 runs averaged).
worst area__x__worst concave points	✓ KEPT	MutualInformation	MI = $0.5258 \pm 0.0000 \geq$ threshold 0.0100. (3 runs averaged).
worst texture	✓ KEPT	StatisticalTest	ANOVA F-test: $F=127.37$, $p=3.36e-26$ (significant at $\alpha=0.0031$ with Bonferroni).
worst perimeter	✓ KEPT	StatisticalTest	ANOVA F-test: $F=767.90$, $p=1.40e-99$ (significant at $\alpha=0.0031$ with Bonferroni).
category_col	✓ KEPT	StatisticalTest	ANOVA F-test: $F=9339.57$, $p=1.79e-304$ (significant at $\alpha=0.0031$ with Bonferroni).
signup_date_year	✓ KEPT	StatisticalTest	ANOVA F-test: $F=16.66$, $p=5.28e-05$ (significant at $\alpha=0.0031$ with Bonferroni).
signup_date_month_sin	X DROPPED	StatisticalTest	ANOVA F-test: $F=0.29$, $p=5.89e-01$ (not significant at $\alpha=0.0031$ with Bonferroni).
signup_date_month_cos	X DROPPED	StatisticalTest	ANOVA F-test: $F=0.54$, $p=4.62e-01$ (not significant at $\alpha=0.0031$ with Bonferroni).
signup_date_dow_sin	X DROPPED	StatisticalTest	ANOVA F-test: $F=0.02$, $p=8.79e-01$ (not significant at $\alpha=0.0031$ with Bonferroni).
signup_date_day	X DROPPED	StatisticalTest	ANOVA F-test: $F=1.65$, $p=1.99e-01$ (not significant at $\alpha=0.0031$ with Bonferroni).
signup_date_quarter	X DROPPED	StatisticalTest	ANOVA F-test: $F=3.77$, $p=5.29e-02$ (not significant at $\alpha=0.0031$ with Bonferroni).
signup_date_days_since_epoch	✓ KEPT	StatisticalTest	ANOVA F-test: $F=36.85$, $p=2.70e-09$ (significant at $\alpha=0.0031$ with Bonferroni).
category_col__x__mean area	✓ KEPT	StatisticalTest	ANOVA F-test: $F=815.54$, $p=2.38e-103$ (significant at $\alpha=0.0031$ with Bonferroni).
category_col__x__area error	✓ KEPT	StatisticalTest	ANOVA F-test: $F=3272.49$, $p=2.16e-209$ (significant at $\alpha=0.0031$ with Bonferroni).
category_col__x__worst concavity	✓ KEPT	StatisticalTest	ANOVA F-test: $F=133.97$, $p=2.54e-27$ (significant at $\alpha=0.0031$ with Bonferroni).
worst perimeter__x__worst concavity	✓ KEPT	StatisticalTest	ANOVA F-test: $F=627.63$, $p=1.48e-87$ (significant at $\alpha=0.0031$ with Bonferroni).
worst perimeter__x__radius error	✓ KEPT	StatisticalTest	ANOVA F-test: $F=496.55$, $p=8.17e-75$ (significant at $\alpha=0.0031$ with Bonferroni).
worst area__x__worst concave points	✓ KEPT	StatisticalTest	ANOVA F-test: $F=800.53$, $p=3.54e-102$ (significant at $\alpha=0.0031$ with Bonferroni).

Feature	Final Status	Pipeline Step	Action & Reasoning
worst texture	X DROPPED	RFE	RFE Rank 7. Eliminated: low tree-based feature importance.
worst perimeter	X DROPPED	RFE	RFE Rank 4. Eliminated: low tree-based feature importance.
category_col	✓ KEPT	RFE	RFE Rank 1 (selected, top-tier feature importance).
signup_date_year	X DROPPED	RFE	RFE Rank 8. Eliminated: low tree-based feature importance.
signup_date_days_since_epoch	X DROPPED	RFE	RFE Rank 5. Eliminated: low tree-based feature importance.
category_col__x__mean area	✓ KEPT	RFE	RFE Rank 1 (selected, top-tier feature importance).
category_col__x__area error	✓ KEPT	RFE	RFE Rank 1 (selected, top-tier feature importance).
category_col__x__worst concavity	X DROPPED	RFE	RFE Rank 2. Eliminated: low tree-based feature importance.
worst perimeter__x__worst concavity	X DROPPED	RFE	RFE Rank 3. Eliminated: low tree-based feature importance.
worst perimeter__x__radius error	X DROPPED	RFE	RFE Rank 6. Eliminated: low tree-based feature importance.
worst area__x__worst concave points	✓ KEPT	RFE	RFE Rank 1 (selected, top-tier feature importance).